
RailEase

IRCTC Website UI/UX Redesign

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INTRODUCTION

RailEase is a modern UI/UX redesign project created to improve the usability and visual experience of the existing railway ticket booking platform Indian Railway Catering and Tourism Corporation (IRCTC). The current website contains complex layouts, crowded interfaces, and outdated visual elements that may create difficulties for users during the ticket booking process.

This project focuses on redesigning the website with a clean, modern, and user-friendly interface inspired by contemporary travel booking platforms. The redesign aims to simplify navigation, improve accessibility, enhance visual hierarchy, and create a smoother booking experience for railway passengers.

The redesigned system includes important user flows such as train search, train details, passenger information, payment processing, booking confirmation, and booking management. High-fidelity mockups were created using Figma while maintaining consistency in colors, typography, spacing, and responsive layout principles.

The project also applies UI/UX design concepts such as heuristic evaluation, mood boards, style guides, and user-centered design approaches to create an efficient and visually appealing railway booking experience.

Existing Website Problems

The existing railway booking website contains several usability and interface-related issues that affect the overall user experience. The interface appears crowded with excessive information, multiple menus, and inconsistent visual hierarchy, making navigation difficult for users.

One of the major problems identified is the outdated design style. The website uses old layout structures, limited spacing, and cluttered sections, which reduce readability and create confusion during ticket booking. Important actions such as train search, seat selection, and payment are not visually highlighted effectively.

The booking process also requires users to navigate through multiple complex sections, increasing the cognitive load and reducing usability. The lack of modern UI elements, responsive layouts, and

intuitive navigation negatively impacts both desktop and mobile users.

To solve these issues, the RailEase redesign project focuses on improving navigation flow, simplifying booking interactions, enhancing visual hierarchy, and creating a clean, modern, and user-friendly interface with better accessibility and responsiveness.

Heuristic Evaluation

A heuristic evaluation was performed on the existing railway booking website to identify usability issues and improve the overall user experience. The evaluation was based on common UI/UX usability principles such as consistency, visibility, navigation, error prevention, and user control.

1. Visibility of System Status

The website provides limited visual feedback during ticket booking and navigation processes. Users may find it difficult to understand the current stage of booking or payment progress.

2. Consistency and Standards

Different sections of the website use inconsistent layouts, button styles, font sizes, and spacing. This creates confusion and reduces interface clarity.

3. User Control and Freedom

Users may face difficulty while modifying booking details or navigating back to previous steps during the booking process.

4. Error Prevention

Important fields and actions are not clearly highlighted, increasing the possibility of user mistakes while entering travel information or passenger details.

5. Recognition Rather Than Recall

The interface contains excessive information and crowded menus, forcing users to remember navigation paths and booking steps.

6. Aesthetic and Minimalist Design

The existing design lacks modern UI structure, proper spacing, and visual hierarchy. The interface appears cluttered and visually outdated.

7. Flexibility and Efficiency

The website provides limited personalization and responsive support for modern devices, affecting efficiency for both desktop and mobile users.

Mood Board

The mood board was created to define the overall visual style and design direction of the RailEase railway booking website redesign. The design inspiration was taken from modern travel and airline booking platforms that focus on clean layouts, smooth user experience, and premium visual appearance.

The redesign mainly uses purple, indigo, white, and soft lavender color combinations to create a modern and trustworthy travel experience. Rounded cards, soft shadows, gradient backgrounds, and glassmorphism effects were selected to improve the visual appeal of the interface.

Modern typography styles with proper spacing and minimal clutter were used to enhance readability and navigation. Futuristic train illustrations, modern booking cards, dashboard layouts, and responsive structures were included as part of the visual inspiration.

The mood board helped maintain consistency in colors, typography, button styles, card layouts, and overall user interface design throughout the project.

STYLE GUIDE

The RailEase redesign follows a modern and consistent visual style to improve usability and user experience. A unified style guide was created to maintain consistency across all screens and components.

Color Palette

The interface mainly uses purple, indigo, white, and soft lavender shades to create a clean and premium travel booking experience.

Typography

Modern fonts with clear readability and proper spacing were used for headings, buttons, and body text to improve visual hierarchy.

Buttons

Rounded buttons with consistent colors and hover styles were designed to highlight important user actions such as booking tickets and making payments.

Cards and Layouts

Soft shadows, rounded corners, and spacious layouts were used to reduce clutter and improve content organization.

HIGH-FIDELITY MOCKUPS

High-fidelity mockups of the redesigned railway booking website were created using Figma. The mockups were designed with modern UI/UX principles to improve usability, navigation flow, and visual appearance.

The redesign includes multiple interconnected screens such as the Home Page, Train Search Results, Train Details, Passenger Details, Payment Page, Booking Confirmation, and My Bookings dashboard.

Each screen was designed with responsive layouts, clean spacing, modern typography, rounded cards, soft shadows, and consistent color usage. The interface focuses on reducing visual clutter and creating a smoother booking experience for users.

Interactive components and organized layouts were used to simulate a realistic railway booking platform with improved accessibility and user-friendly navigation.

RESPONSIVE LAYOUT

The RailEase website redesign was created using responsive layout principles to provide a smooth and user-friendly experience across different devices and screen sizes. The interface was designed to adapt properly for desktop, tablet, and mobile layouts while maintaining consistency in design and usability.

Flexible grid structures, spacious layouts, scalable cards, and organized content alignment were used to improve readability and accessibility. Navigation menus, booking sections, buttons, and information cards were arranged carefully to ensure easy interaction for users on different screen resolutions.

The responsive structure also helps reduce visual clutter and improves the overall booking experience by maintaining proper spacing, typography hierarchy, and component consistency throughout the website. Modern UI elements and adaptable layouts were implemented to create a clean and efficient railway booking platform.

CONCLUSION

The RailEase UI/UX redesign project successfully improved the visual appearance, usability, and overall user experience of the existing railway booking website. The redesigned interface focuses on clean layouts, modern design elements, better navigation flow, and responsive structures to create a smoother booking experience for users.

Through heuristic evaluation, mood board planning, style guide creation, and high-fidelity mockups, the project demonstrated the importance of user-centered design principles in modern web applications. The redesigned system simplifies the railway ticket booking process while improving accessibility, readability, and visual hierarchy.

This project provided practical experience in UI/UX design, responsive web design, visual hierarchy, and modern interface planning using Figma and GitHub.